Work in progress — draft, not peer reviewed

The Frohmanian Tethered Metriplectic World Model in Lean 4: Canonical Bracket Splitting, Encoding Obstructions, and the Newton Potential

Companion paper to `BenFrohman/metriplectic-world-model`

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Draft — tracks build in progress. This repository formalizes a *model*: the tethered metriplectic system, which adds a symmetric tether bracket and cubic damping of peak vorticity to 3D vorticity dynamics. It proves nothing about unmodified Navier–Stokes ($\kappa = 0$) and does not resolve the Clay Millennium problem.

Abstract

We report a Lean 4/Mathlib [2, 4] formalization (0 `sorry`; axioms only `propext`, `Classical.choice`, `Quot.sound`) of the author's tethered metriplectic model (in the sense of [3]). Proved unconditionally: the unique antisymmetric/symmetric splitting of a bilinear bracket (`split_exists_unique`); a non-circular canonicity theorem forcing the metric bracket to the form $c\langle \Pi_u x, \Pi_u y\rangle$ (`canonical_metric_bracket`); a uniform vorticity ceiling for the comparison inequality $M' \le CM^2 - \kappa M^3$; a log-loss replacement for the stretching hypothesis giving a ceiling when $\kappa > 4C_{\mathrm{abs}}$; several encoding counterexamples; the Newton-potential identity $\Delta(N \star g) = -4\pi g$ on $\mathbb{R}^3$ for $g \in C_c^2$; and $\operatorname{curl} \mathrm{BS}\,\omega = \omega$ for smooth, compactly supported, divergence-free ω (`curlBiotSavartTest`). Conditional and open items are listed by their exact Prop names.

2020 Mathematics Subject Classification. 35Q30, 76D05, 68V15; 37K05, 76B47, 03B35.
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1 Introduction

Metriplectic dynamics [3] combine a Hamiltonian (antisymmetric) bracket with a dissipative (symmetric) bracket. This paper documents a Lean 4 formalization of the author's tethered metriplectic model of three-dimensional vorticity dynamics, in which a symmetric tether bracket adds cubic damping of the peak vorticity. We establish the algebraic structure of the brackets (Section 4), the comparison-ODE ceilings and their dependence on the stretching hypothesis (Section 5), several obstructions arising from the encoding (Section 6), and the analytic facts about the Newton potential and the Biot–Savart law on $\mathbb{R}^3$ needed for the helicity results. The model is not the Navier–Stokes system, and no statement about unmodified Navier–Stokes is made.

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2 Naming

`Pi_u` is the author's own name (manuscript of 31 May 2026): the projection off span$\{u\}$,

$$\Pi_u x = x - \frac{\langle u, x\rangle}{\|u\|^2}\, u. \tag{2.1}$$

All other identifiers defined in this repository are *coined in this build* (marked †), including `canonical_metric_bracket`, kept by the author's choice and flagged as coined. Modules under `FrohmanianSymplecticTether/` and `TetherMayForm/` are ported verbatim from earlier repositories.

3 Manuscript framing

The May/June 2026 manuscript introduces a tethered Lyapunov functional S_ε and reduces regularity to the comparison ODE

$$y' = Cy^2 - \kappa'' y^3 \tag{3.1}$$

for the peak vorticity, through transport cancellation, viscous dissipation at the maximum, a Hölder–Sobolev gate, and Young absorption at $\varepsilon = \kappa/4$ (Proposition 2). Within the model, the cubic term comes from the tether. The manuscript's hypothesis (H5), a non-circular majorant for the vorticity, is the genuine gap; for $\kappa = 0$ it is the Clay core. Here κ has units of time and breaks Navier–Stokes scaling, which is why the model is distinct from NS. *Normalization of κ:* where the Lean code fixes $\kappa = 3/2$ (`CalderonZygmundConstant3D`, defined as $(3/(8\pi))\cdot 4\pi$), this is a convention; it is not a computed Calderón–Zygmund operator norm and not derived from a Conley–Zehnder index. The ceilings hold for every $\kappa > 0$.

4 Brackets and canonicity (`Bracket.lean`, `Canonical.lean`)

Theorem 4.1. `split_exists_unique`†*: every bilinear bracket is uniquely an antisymmetric part plus a symmetric part.*

Theorem 4.2. `canonical_metric_bracket`†*: on a real inner product space, a bilinear form B that is symmetric (C1), degenerate along u (C2: $B(u,\cdot) = 0$) and invariant under every linear isometry fixing u equals*

$$B(x, y) = c\,\langle \Pi_u x, \Pi_u y\rangle.$$

The hypotheses do not mention Π_u. `canonical_metric_bracket_sign`†*: if $B(x,x) \le 0$ then $c \le 0$.* `Pi_u_form_symm_degen`†*: the forced form satisfies C1, C2.*

Scope: pointwise at one state; does not force the $|\omega|^2$ weight, the value of κ, or C3. Also proved: `energy_conserved`, `energy_const_along`, `entropy_production`, `gen_entropy_eq`, `entropy_monotone_along`†.

5 Ceiling and stretching (`Ceiling.lean`, `AuditFixes.lean`)

Theorem 5.1. `uniform_ceiling`†*: if $M' \le CM^2 - \kappa M^3$ and $\kappa > 0$, then*

$$M(t) \le \max\big(M(0), \max(C/\kappa, 0)\big) \qquad (t \ge 0). \tag{5.1}$$

Also `absorbing`, `ceiling_antitone_kappa`, `kappa_zero_no_ceiling`† *(at $\kappa = 0$, $1/(1-t)$ satisfies $M' \le M^2$ and blows up).*

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Correction. The hypothesis `StretchingCZ` (a pure y^3 bound on stretching at the maximum) is *false as a sup-norm bound* for Navier–Stokes: the known bound carries a logarithmic loss. Results using it are relabeled as model-only. Its replacement, consistent with the known estimates in form, is `StretchingLogLoss` $d\,C_1\,C_2$†:

$$St \le C_1 y^3 + C_2 y^3 \log(1+y).$$

Theorem 5.2. `maxpoint_rate_logLoss`†*: under* `EnergyIdentity`, `StretchingLogLoss`, `ViscousAtMax`, `TetherFeedback`, `CrossControlled`,

$$y' \le \Big(C_1 + \tfrac{\kappa}{4} + \tfrac{4C_2^2}{\kappa}\Big) y^2 - \Big(\tfrac{\kappa}{4} - C_{\mathrm{abs}}\Big) y^3,$$

so with `uniform_ceiling` *a uniform ceiling follows whenever* $\kappa > 4C_{\mathrm{abs}}$ *(Young absorption at* $\varepsilon = \kappa/4$*).*

6 Encoding obstructions (`ChoiceFree`, `HelmholtzObstruction`, `TestClass`)

- `not_streamFunctionRepModGrad`†: `StreamFunctionRepModGrad` fails for every ω — representatives are determined only a.e.; uniqueness must be stated almost everywhere (open).
- `arnoldBracketOf_antisymm`, `entropyCasimir_arnold`†: a choice-free Arnold bracket [1] and a Casimir criterion.
- `not_curl_biotSavart_eq`†: curl(BiotSavart ω) $= \omega$ fails for the point spike in the current encoding; `helicity_casimir_of_curlBiotSavart`† is conditional on `CurlBiotSavartOn`.
- `IsTestOrbit`† (smooth, compactly supported, div-free); `not_isTestOrbit_pointSpike`, `helicity_casimir_test`† (conditional on `CurlBiotSavartTest`, discharged in `helicity_casimir_test_closed`, Section 7); `integrableOn_ball_norm_rpow_neg`†.

7 Newton potential on $\mathbb{R}^3$ (`NewtonKernel`, `RegularizedKernel`, `NewtonLaplacian`)

Proved: `scalarLaplacian_newtonRadial` ($\Delta(1/|y|) = 0$, $y \neq 0$), `contDiff_newtonPotential`, `hasFDerivAt_newtonPotential` (differentiation under the integral); for $N_\varepsilon = (|y|^2+\varepsilon^2)^{-1/2}$: `scalarLaplacian_regNewton` ($\Delta N_\varepsilon = -3\varepsilon^2(|y|^2+\varepsilon^2)^{-5/2}$), `regNewton_le`, `tendsto_regNewton`, `integral_neg_scalarLaplacian_regNewton` ($\int -\Delta N_\varepsilon = 4\pi$), `tendsto_regPeak_approxIdentity` (all †). The regularized route avoids spherical Green identities, which Mathlib lacks. Step 3 is closed (`NewtonLaplacian`): `integral_mul_scalarLaplacian_eq` ($\int K\,\Delta g = \int \Delta K\, g$ for $K \in C^2$, $g \in C_c^2$, by Mathlib integration by parts, no boundary terms), `regNewton_conv_laplacian` ($N_\varepsilon \star \Delta g = (\Delta N_\varepsilon) \star g$), `tendsto_regNewton_conv` (the $\varepsilon \to 0$ limit, dominated by $N|h|$), `scalarLaplacian_newtonPotential` ($\Delta(N \star g) = N \star \Delta g$), and

Theorem 7.1. `scalarLaplacian_newtonPotential_eq`†*: for* $g \in C_c^2(\mathbb{R}^3)$ *and* $N(y) = 1/|y|$, $\Delta(N \star g)(x) = -4\pi g(x)$ *for every* x.

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8 Biot–Savart on the test class (`BiotSavartNewton`)

With $G_j(t) = t_j/|t|^3$ (`gradKer`† at $\varepsilon = 0$):

Theorem 8.1. `newtonPotential_deriv_eq`†*: for* $h \in C^1_c$, $N \star \partial_j h = -\int G_j(t)\, h(x-t)\, dt$. *(Regularized identity* `regNewton_conv_deriv`*, then* $\varepsilon \to 0$ *on both sides; the right side is dominated by* $|t|^{-2}|h|$*.)*

Theorem 8.2 (Step 5)**.** `biotSavart_eq_curl_newtonVec`†*: for* `IsTestOrbit` ω, $\mathrm{BiotSavart}\,\omega = (4\pi)^{-1}\,\mathrm{curl}(N \star \omega)$ *pointwise,* $N \star \omega$ *taken componentwise (*`newtonVec`*).*

Theorem 8.3 (Step 4)**.** `curl_curl_newtonVec`†*: for* `IsTestOrbit` ω, $\mathrm{curl}\,\mathrm{curl}(N \star \omega) = 4\pi\omega$, *using symmetry of second derivatives (*`fderiv_fderiv_swap`*),* $\mathrm{div}(N \star \omega) = N \star \mathrm{div}\,\omega = 0$, *and the step-3 theorem.*

Theorem 8.4. `curlBiotSavartTest`†*:* `CurlBiotSavartTest` *holds, i.e.* $\mathrm{curl}\,\mathrm{BiotSavart}\,\omega = \omega$ *for every smooth, compactly supported, divergence-free* ω*. Consequently* `helicity_casimir_test_closed`†*: helicity is a Casimir of the Arnold bracket at such* ω*, given the explicit argument* $D\,S = u$*.*

Scope: this is the classical identity on the smooth compactly supported class only; it says nothing about the dynamics. The Gâteaux derivative of helicity is treated in Section 9.

9 Helicity on the test class (`HelicityGateaux`)

Correction to a definition. `IsSmoothOrbit` previously used `ContDiff` $\mathbb{R}$ $\top$, which in current Mathlib means analytic; a compactly supported test orbit was then 0 and the earlier test-class statements were vacuous. It now uses C^∞ (the order $(\top : \mathbb{N}_\infty)$ coerced to `WithTop` $\mathbb{N}_\infty$); the class contains every $\mathrm{curl}\,\psi$ with $\psi \in C^\infty_c$ (`isTestOrbit_curlOrbit`†).

Theorem 9.1. `biotSavart_selfAdjoint`†*: for continuous compactly supported* a, b, $\int \langle Ba, b\rangle = \int \langle Bb, a\rangle$ *(Fubini with the majorant* $|x-y|^{-2}|a(y)||b(x)|$*, kernel symmetry, triple product).*

Theorem 9.2. `helicity_gateaux`†*: with* $\mathcal{H}(\omega) = \int \langle B\omega, \omega\rangle$ *(*`helicity`†*),* $2B\omega$ *is a Gâteaux representative of* $\mathcal{H}$ *on the test class.*

Theorem 9.3. `curl_eq_of_isGateauxRepOn`†*: two* C^1 *representatives of the same functional on the test class have equal curl. Hence* `helicity_casimir_of_rep`†*: if* $D\,S$ *is any* C^1 *representative of* $\mathcal{H}$*, then* $\mathrm{curl}\,D\,S = 2\omega$ *and* S *is a Casimir of the Arnold bracket* `ArnoldBracketOf`*; no hypothesis on the value of* $D\,S$*.*

Theorem 9.4. `streamFunctionRepModGradOn_of_exists`†*:* `StreamFunctionRepModGradOn` *follows from one* C^2 *kinetic-energy representative* ψ *with* $\mathrm{curl}\,\psi = u$ *and the Poincaré lemma* `CurlFreeIsGradient`†*; the curl half of uniqueness is* `kineticEnergy_reps_curl_eq`†*.*

10 Poincaré lemma and the kinetic-energy representative (`PoincareR3`, `KineticEnergyRep`)

Theorem 10.1. `curlFreeIsGradient`†*: every* C^2 *curl-free field on* $\mathbb{R}^3$ *is the gradient of a* C^2 *function.*

Theorem 10.2. `decays_newtonPotential`†*: if* g *is continuous, compactly supported and* $\int g = 0$*, then* $|N \star g(x)| \le C(1+|x|)^{-2}$*. Derivatives of* C^1_c *functions and the coordinates of a divergence-free test field have mean zero (*`IsTestOrbit.integral_coord_eq_zero`†*).*

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Theorem 10.3. `integral_pairing_curl_symm`†*:*

$$\int \langle a, \operatorname{curl} b\rangle \, \mathrm{d}x = \int \langle \operatorname{curl} a, b\rangle \, \mathrm{d}x$$

for fields with decaying coordinates and first derivatives. Hence, with $\psi = (4\pi)^{-1} N \star \omega$ *(*`keStream`†*),* $\int \langle B\omega, B\varepsilon\rangle = \int \langle \psi, \varepsilon\rangle$*, so* ψ *is a* C^∞ *Gâteaux representative of the kinetic energy with* $\operatorname{curl}\psi = u$ *(*`keStream_isGateauxRep`†*).*

Theorem 10.4. `streamFunctionRepModGradOn`†*:* `StreamFunctionRepModGradOn` *holds unconditionally on the test class.*

Smoothness audit. `instSmooth` in `Instance.lean` stated analyticity where C^∞ was meant (still true, now stated as C^∞). `contDiffAt_newtonRadial` states analyticity of $1/|x|$ off 0; that is true and proved, so it is left as is.

11 Maximizer data from a vorticity field (`MaxPointDanskin`, `EntropyCasimirArnold`)

Theorem 11.1. `danskin_upper_dini`†*: if* $f, \partial_t f$ *are jointly continuous on* $\mathbb{R} \times K$*,* K *compact, then for some maximizer* $x_\star$ *of* $f(t,\cdot)$ *on* K *and every* $\varepsilon > 0$*,* $\sup_K f(t+h) \le f(t,x_\star) + h(\partial_t f(t,x_\star) + \varepsilon)$ *for small* $h > 0$*; i.e.* $D^+ \sup_K f(t) \le \partial_t f(t,x_\star)$*.* `vorticity_danskin_upper_dini`† *applies it to* $|\omega|^2$*.*

Theorem 11.2. `sup_mul_deriv_eq_inner`†*: if* $Y = \sup|\omega|$ *is differentiable at* t*, then* $YY' = \omega \cdot \partial_t \omega$ *at every global maximizer. At such a point* $\omega \cdot (v \cdot \nabla)\omega = 0$ *(*`inner_convective_eq_zero_of_max`†*) and* $\omega \cdot \Delta\omega \le 0$ *for* C^2 *fields (*`inner_laplacian_nonpos_of_max`†*).*

Theorem 11.3. `maxPointRealization_of_field`†*: for a solution of* `ModelVorticityPDE` *with* $\nu \ge 0$*,* $\omega(t) \in C^2_c$*,* C^1 *in time, and* $Y = \sup|\omega| > 0$ *differentiable,* `MaxPointRealization` *holds with* `EnergyIdentity`*,* `ViscousAtMax` *and* `CrossControlled` *discharged. The remaining hypotheses are* `FieldStretchingBound`† *and* `FieldTetherFeedback`†*. Running the ceiling argument from the one-sided Dini bound alone is open.*

Theorem 11.4. `arnoldCasimirField_helicity`†*:* $2u$ *satisfies* `ArnoldCasimirField`† *(*$\forall a$*,* `ArnoldBracketOf` $\omega\, a\, (2u) = 0$*) on the test class. This is the clean form of* `EntropyCasimir` *for the concrete bracket (*`entropyCasimir_of_arnoldCasimirField`†*). The choice-based form* `EntropyCasimirChoice`† *is kept and marked not provable as stated.*

12 The ceiling from the Dini bound (`DiniCeiling`)

Theorem 12.1. `ceiling_of_upper_dini`†*: if* M *is continuous on* $[0,\infty)$*,* $M(t+h) \le M(t) + h(R(t) + \varepsilon)$ *for small* $h > 0$*, and* $R(t) < 0$ *whenever* $M(t) > B$*, then* $M(0) \le B$ *implies* $M \le B$ *on* $[0,\infty)$*.*

Theorem 12.2. `field_uniform_ceiling`†*: take a solution of* `ModelVorticityPDE` *with* $\nu \ge 0$*,* $\kappa > 0$*,* C^2 *in space,* C^1 *in time, and support in a fixed compact set. Under* `FieldStretchingBound` *and* `FieldTetherFeedback`*,* $\sup|\omega(t)| \le \max(\sup|\omega(0)|, C/\kappa, 0)$ *for all* $t \ge 0$*. No differentiability of* $\sup|\omega|$ *is assumed; its continuity is proved (*`sup_continuous_of_uniform_support`†*).*

Theorem 12.3. `fieldTetherFeedback_cubicTether`†*: for* $T = -\kappa|\omega|^2\omega$ *(*`cubicTether`†*),* `FieldTetherFeedback` *holds. Hence* `field_uniform_ceiling_cubicTether`† *needs only* `FieldStretchingBound`*. For a general tether it does not follow.*

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13 Log-loss stretching with the cubic tether (`LogLossCeiling`)

Theorem 13.1. `field_uniform_ceiling_logLoss_cubicTether`†*: take a solution of* `ModelVorticityPDE` *with the cubic tether, under the regularity and uniform-support hypotheses of* `field_uniform_ceiling`*. Assume the named Prop* `FieldStretchingLogLoss`†*,* $\omega\cdot(\omega\cdot\nabla)u \le C_1Y^3 + C_2Y^3\log(1+Y)$ *at maxima. Then* $\sup|\omega(t)| \le \mathrm{ceiling}(Y_0,\, C_1+\kappa/4+4C_2^2/\kappa,\, 3\kappa/4)$ *for every* $\kappa > 0$.

This is a reduction, not an unconditional result. The Calderón–Zygmund estimate is not formalized here. Its known form has $\log^+\big(\|\omega\|_{C^\alpha}/Y\big)$, a higher norm that is not controlled uniformly in time, so it would not close the argument. The Y-only log loss is a strictly stronger assumption.

14 Status

- **Proved unconditionally:** `split_exists_unique`, `canonical_metric_bracket` (+ sign, symm/degen), energy/entropy structure, `uniform_ceiling`, `maxpoint_rate_logLoss` (as an implication from its listed Props), the obstructions above, the Newton potential identity (steps 1–3), `curlBiotSavartTest` (steps 4–5), Biot–Savart self-adjointness, the Poincaré lemma `curlFreeIsGradient`, `streamFunctionRepModGradOn` on the test class, $D\mathcal{H} = 2u$ and the helicity Casimir of the concrete Arnold bracket on the test class (`helicity_casimir_of_rep`), the scalar instance `instance_global_regularity`, the Danskin bound `danskin_upper_dini` and the maximum-principle lemmas.
- **Conditional on named Props:** `model_uniform_ceiling` (`TetheredMaxRate`); `model_global_regularity` (`TetheredMaxRate`, `ModelContinuation`); `model_global_regularity_of_realization` (`MaxPointRealization`, whose fields include `StretchingCZ` — false as an L^∞ NS bound); velocity-tether energy/entropy (`KineticEnergyCurlIdentity`, `EntropyCasimir`); helicity Casimir on the test class given $D\,S = u$ (`helicity_casimir_test_closed`; no longer conditional on `CurlBiotSavartTest`).
- **Open:** `EntropyCasimir` for the choice-based `FunctionalDerivative` bracket (not provable as stated); deriving the $|\omega|^2$ weight, κ and C3 from C1–C3; for the field-level realization (`maxPointRealization_of_field`): `FieldStretchingBound` (or `FieldStretchingLogLoss`; the CZ estimate is not formalized, and its C^α form would not close the ceiling), `FieldTetherFeedback` for tethers other than `cubicTether`, and existence and regularity (uniform compact support) of actual solutions. Differentiability of $\sup|\omega|$ is no longer needed (`field_uniform_ceiling`); and, for unmodified NS, the (H5) majorant, i.e. the Clay core. Neither the Clay problem nor NS regularity is claimed.
- The type named `T3` is $\mathbb{R}^3$, not the torus.

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